THE STALK OF THE PUSHFORWARD OF A CONSTANT SET ALONG A POINT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; every step the author writes out is inside the counted span and no closing sentence is a placeholder.

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Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- The adjunction between the pullback and the pushforward of a point. Same chapter, the adjunction defining a point of a topos. Supplies the map from the stalk of the pushforward to the set.
- Exactness of the functors attached to a point. Same chapter, the fact that the pullback and pushforward of a morphism of topoi are left exact. Shows that the final object is preserved, which is the base case of the argument.

Lemma 0.1. *Let $\mathcal{C}$ be a site. Let $p : Sh(pt) \to Sh(\mathcal{C})$ be a point of the topos associated to $\mathcal{C}$. For any set E there are canonical maps*

$$E \longrightarrow (p_*E)_p \longrightarrow E$$

whose composition is id_E.

Proof. There is always an adjunction map $(p_*E)_p = p^{-1}p_*E \to E$. This map is an isomorphism when $E = \{*\}$ because p_* and p^{-1} are both left exact, hence transform the final object into the final object. Hence given $e \in E$ we can consider the map $i_e : \{*\} \to E$ which gives

$$\begin{array}{ccc} p^{-1}p_*\{*\} & \xrightarrow{p^{-1}p_*i_e} & p^{-1}p_*E \\ \cong\downarrow & & \downarrow \\ \{*\} & \xrightarrow{i_e} & E \end{array}$$

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whence the map $E \to (p_*E)_p = p^{-1}p_*E$ as desired. $\square$